

Roles-Lab

- Create a web role, and ensure that `/var/www/html/index.html` contains the Ansible hostname of the server and all IPv6 addresses connected to that server. Ensure that `httpd` is running and enabled and that the role deploys correctly.

Answer:

First Create ansible role directory using

- `Ansible-galaxy init <Role Name>` → `ansible-galaxy init Web_Role`

```
[root@ControlledNode Roles]# tree
```

```
.
├── Web-Role
│   ├── README.md
│   ├── defaults
│   │   └── main.yml
│   ├── files
│   ├── handlers
│   │   └── main.yml
│   ├── meta
│   │   └── main.yml
│   ├── tasks
│   │   └── main.yml
│   ├── templates
│   │   └── index.j2
│   ├── tests
│   │   ├── inventory
│   │   └── test.yml
│   └── vars
│       └── main.yml
```

```
9 directories, 9 files
```

```
[root@ControlledNode Roles]# |
```

2. Second configure the **ansible.cfg** file to allocate the Role Directory

[defaults]

Roles_path = <path to your Role Directory>

```
[root@ControlledNode 1-Ansible-Intermediate-skill]# cat ansible.cfg
[defaults]
inventory=./inventory
remote_user=ec2-user
private_key_file=Ansibles.pem
ask_pass=false

roles_path=/home/ec2-user/Ansible_Projects/Ansible_Tasks/1-Ansible-Intermediate-skill/Roles/

[privilege_escalation]
become=true
become_method=sudo
become_user=root
```

3. Create your Files inside the Role Directory

- Handlers: for the handlers tasks happen when change the state
- Tasks: for your Playbook tasks
- Vars: for variables [high priority]
- Default: for vars [low priority]
- Templates: for jinja2 files

```
[root@ControlledNode Web-Role]# cat handlers/main.yml
---
# handlers file for Web-Role
- name: restart_httpd
  service:
    name: httpd
    state: restarted
[root@ControlledNode Web-Role]# |
```

```
[root@ControlledNode Web-Role]# cat tasks/main.yml
---
# tasks file for Web-Role
- name: Install Httpd package
  yum:
    name: httpd
    state: latest
- name: Start the Httpd Service
  service:
    name: httpd
    state: started
    notify: restart_httpd
- name: Setting up index.html
  template:
    src: index.j2
    dest: /var/www/html/index.html

[root@ControlledNode Web-Role]# |
```

4. Create the Main playbook to run your Role hint: Create the Main playbook within your ansible.cfg and inventory file

```
[root@ControlledNode Web-Role]# cat ../../Generator_Roles.yml
---
- name: testing roles one by one
  hosts: managednode1
  roles:
    - Web-Role
[root@ControlledNode Web-Role]# |
```

```
[root@ControlledNode 1-Ansible-Intermediate-skill]# ansible-playbook Generator_Roles.yml

PLAY [testing roles one by one] *****

TASK [Gathering Facts] *****
[WARNING]: Platform linux on host managednode1 is using the discovered Python interpreter at /usr/bin/python3.9, but fu
See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
ok: [managednode1]

TASK [Web-Role : Install Httpd package] *****
ok: [managednode1]

TASK [Web-Role : Start the Httpd Service] *****
ok: [managednode1]

TASK [Web-Role : Setting up index.html] *****
changed: [managednode1]

PLAY RECAP *****
managednode1      : ok=4    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

```
[root@ControlledNode 1-Ansible-Intermediate-skill]# ansible managednode1 -m command -a 'curl localhost'
[WARNING]: Platform linux on host managednode1 is using the discovered Python interpreter at /usr/bin/python3
See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more infor
managednode1 | CHANGED | rc=0 >>
You Are Viewing Webpage of managednode1

ip6v6 Adresses of this host are :
['fe80::877:d8ff:fe45:ad45'] % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                             Dload  Upload  Total  Spent    Left  Speed
100  104  100  104    0    0  82018      0 --:--:-- --:--:-- --:--:--  101k
[root@ControlledNode 1-Ansible-Intermediate-skill]# |
```

- Create a requirements.yml file to install several roles from Ansible Galaxy, the roles to be included are geerlingguy.nfs and geerlingguy.git version 3.0.0

```
[root@ControlledNode Roles]# cat requirements.yml
---
roles:
  - name: geerlingguy.nfs

  - name: geerlingguy.git
    version: 3.0.0
```

```
[root@ControlledNode Roles]# ansible-galaxy install -r requirements.yml
Starting galaxy role install process
- downloading role 'nfs', owned by geerlingguy
- downloading role from https://github.com/geerlingguy/ansible-role-nfs/archive/2.1.1.tar.gz
- extracting geerlingguy.nfs to /root/.ansible/roles/geerlingguy.nfs
- geerlingguy.nfs (2.1.1) was installed successfully
- downloading role 'git', owned by geerlingguy
- downloading role from https://github.com/geerlingguy/ansible-role-git/archive/3.0.0.tar.gz
- extracting geerlingguy.git to /root/.ansible/roles/geerlingguy.git
- geerlingguy.git (3.0.0) was installed successfully
```

Configure environment variable to include your directory roles path /home/admin/mylabproject

```
[root@ControlledNode Roles]# export ANSIBLE_ROLES_PATH=/home/ec2-user/Ansible_Projects/Ansible_Tasks/1-Ansible-Intermediate-skill/Roles/
[root@ControlledNode Roles]# |
```

to delete environment Variable : **unset ANSIBLE_ROLES_PATH**

to set environment Variable: **export ANSIBLE_ROLES_PATH**

```
[root@ControlledNode 1-Ansible-Intermediate-skill]# export ANSIBLE_ROLES_PATH=/home/ec2-user/Ansible_Projects/Ansible_Tasks/1-Ansible-Intermediate-skill/Roles/
[root@ControlledNode 1-Ansible-Intermediate-skill]# ansible-galaxy list
# /home/ec2-user/Ansible_Projects/Ansible_Tasks/1-Ansible-Intermediate-skill/Roles
- Web-Role, (unknown version)
[root@ControlledNode 1-Ansible-Intermediate-skill]# unset ANSIBLE_ROLES_PATH
[root@ControlledNode 1-Ansible-Intermediate-skill]# ansible-galaxy list
# /home/ec2-user/Ansible_Projects/Ansible_Tasks/1-Ansible-Intermediate-skill/Roles
- Web-Role, (unknown version)
[root@ControlledNode 1-Ansible-Intermediate-skill]# |
```

